Cambodia ANPR System: Revolutionizing Vehicle Identification

Date: 17/10/2024
Prepare by: Songhieng VAN, AI Engineer
Email: songheingvan@gmail.com
Tel: +8618520575183/+85517865598
Institution: Jinan University, Guangzhou, China
Major: Computer Science

Abstract: *With the rapid urbanization of Cambodia, addressing transportation challenges has become paramount. Automated Number Plate Recognition (ANPR) emerges as a potential solution to this. This document serves as a comprehensive exploration of ANPR, offering a clear overview of its functionality. Beginning with the basic utility of the photo verification feature, the content delves into the importance of timestamp and number plate updates. Mapping, a crucial component of the system, promises benefits such as immediate response mechanisms, resource allocation optimization, and bolstered strategy and planning, further fortifying evidence collection and accountability processes. The paper then expounds on the specific advantages ANPR presents to Cambodia, emphasizing economic impacts, enhanced security, and effective traffic management. A detailed study underscores the practicalities of ANPR implementation in the country, covering aspects from location analysis to camera positioning. As urban landscapes evolve, so does the potential of ANPR. This is evident in its capability to enhance traffic management and safety through advanced violation tracking. Lastly, looking towards the future, the document discusses the scalability of ANPR in Cambodia, emphasizing integration with advanced AI and real-time traffic management. In summation, this exposition highlights the transformative potential of ANPR in reshaping Cambodia's transportation landscape for efficiency, safety, and sustainability.*

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Problem Statement: Navigating Cambodia's Transportation Conundrum

Cambodia, with its rich tapestry of tradition and modernity, is undergoing rapid socio-economic transformations. Central to these changes is the unprecedented growth in its transportation sector. In just two decades, the nation has seen an almost three-fold surge in vehicle registrations, with 2022 alone marking an impressive count of 6.7 million vehicles on its roads. Such a meteoric rise, while indicative of progress, has also precipitated infrastructural challenges. Although the roads have expanded, their capacity has struggled to match the vehicular explosion, leading to frequent congestion and commute delays, especially in bustling urban centers like Phnom Penh.

But the challenges aren't solely infrastructural. The current mechanisms for vehicle identification and monitoring in Cambodia rely predominantly on manual labor, which, by nature, is limited in speed and accuracy. Human observers, even with the best of training, can succumb to fatigue and distractions, making them prone to oversights. Such errors, though seemingly innocuous, can have cascading effects, especially in areas demanding meticulous accuracy like security and law enforcement.

Furthermore, as we transition into an age of heightened global security awareness, the trajectory of an unmonitored vehicle can unveil more than its destination. The gaps in Cambodia's surveillance infrastructure mean that vehicles can operate under the radar, posing risks that range from minor infractions to significant security threats. In essence, while Cambodia's growth story in transportation is commendable, it's also interspersed with challenges that need immediate and innovative solutions, like the advent of technologies such as the Automatic Number Plate Recognition (ANPR) system.

Solution Overview: The Advent of the ANPR in Cambodia

The Automatic Number Plate Recognition (ANPR) system is set to revolutionize the transportation and security sectors in Cambodia. Here's a detailed breakdown of the multifaceted advantages this technology brings to the table:

1. Digital Transformation of Infrastructure Monitoring:
- Rapid Data Processing: The ANPR system can process thousands of vehicle license plates in a fraction of the time it would take a human to identify even one. This translates to real-time data acquisition, even during peak traffic hours.
- Scalability: Regardless of the rapid increase in vehicular volume, ANPR systems can be seamlessly scaled up by integrating more cameras, without necessitating a proportional increase in human resources.
- Optimized Traffic Flow: By identifying vehicles that flout traffic rules, are involved in congestion-causing incidents, or are not roadworthy, the ANPR can aid traffic management authorities in implementing effective remedial measures swiftly.

2. Overcoming Manual Limitations:
- Accuracy and Precision: By leveraging advanced machine learning models like YOLOv8, the ANPR system minimizes human error, ensuring accurate vehicle identification consistently.
- Cost-Effective: In the long run, investing in ANPR technology can prove more cost-effective than recurring expenses tied to manual monitoring.

- 24/7 Monitoring: Unlike human operators, ANPR systems can work tirelessly around the clock, ensuring continuous and consistent monitoring.

3. Enhancing National Safety and Security:
- Instant Alerts: Any vehicle identified as stolen, unregistered, or involved in criminal activities can be instantly flagged by the ANPR system. This ensures immediate alerts to the concerned authorities for swift action.
- Database Integration: The ANPR system's integration with national databases allows for real-time cross-referencing. This aids in instant verification of a vehicle's legitimacy and background checks.
- Crime Deterrence: The mere presence and awareness of a state-of-the-art ANPR system can act as a deterrent for criminal activities, as potential wrongdoers know they are more likely to be identified and caught.
- Evidence Collection: In case of incidents or disputes, the data recorded by the ANPR can serve as unbiased, concrete evidence, aiding judicial processes.

4. Future Expansion and Integration:
- Smart City Integration: As Cambodia marches towards a digital future, the ANPR system can be integrated into broader smart city solutions, optimizing everything from traffic light patterns to parking solutions based on real-time traffic data.
- Data-Driven Policies: The extensive data collected by the ANPR can be used by policymakers to make informed decisions, be it related to road infrastructure development, transportation policies, or urban planning.

Technical Cornerstone:

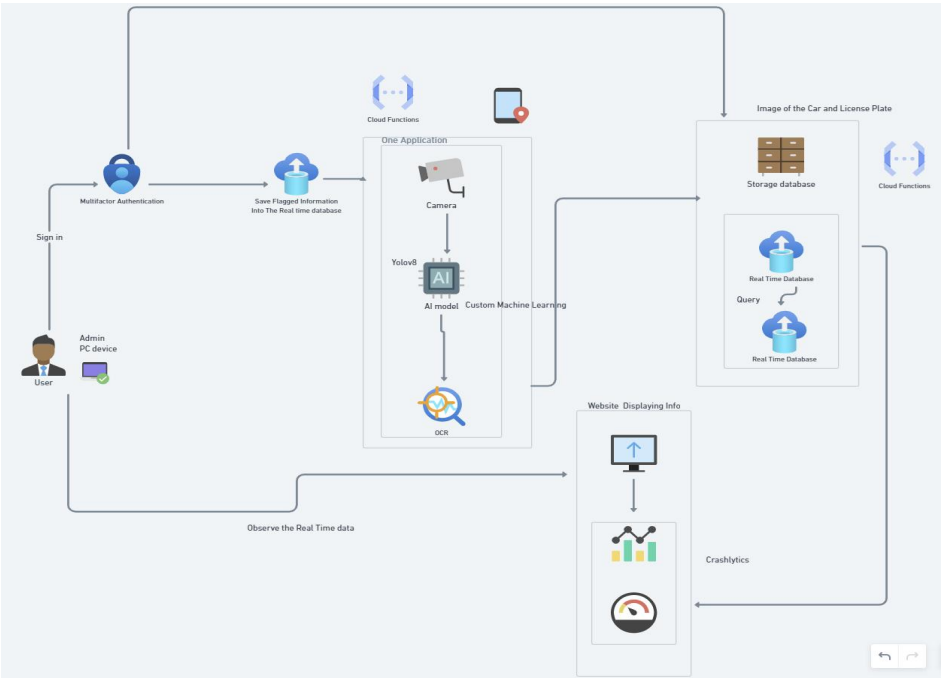


Figure: Architecture of the application

ANPR Machine Learning Pipeline with YOLOv8, Roboflow, and Paperspace:

Blending Roboflow, YOLOv8, and Paperspace, this pipeline exemplifies a modern and efficient ANPR system, promising rapid and accurate license plate detection for Cambodia.

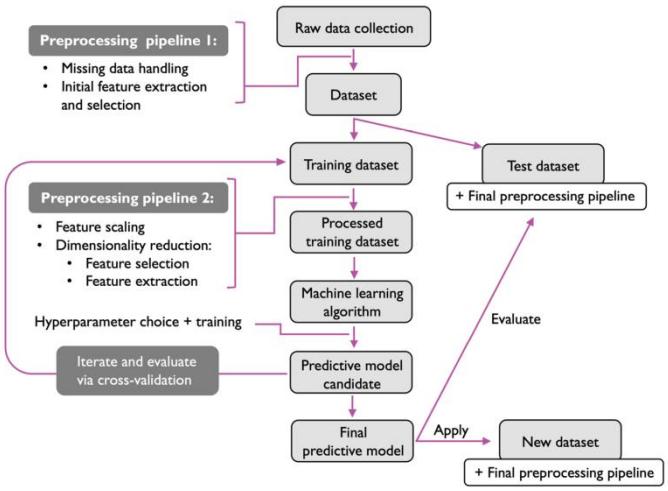


Figure: Diagram Flow of process of the AI model

1. Raw Data Collection:
- Begin by gathering raw image data from cameras throughout Cambodia, focusing primarily on capturing images of vehicle license plates under a variety of conditions.

2. Dataset Formation with Roboflow:
- Utilize Roboflow to collate the raw images into a cohesive dataset. The dataset should consist of labeled data, where each license plate image is paired with its correct alphanumeric license plate value.

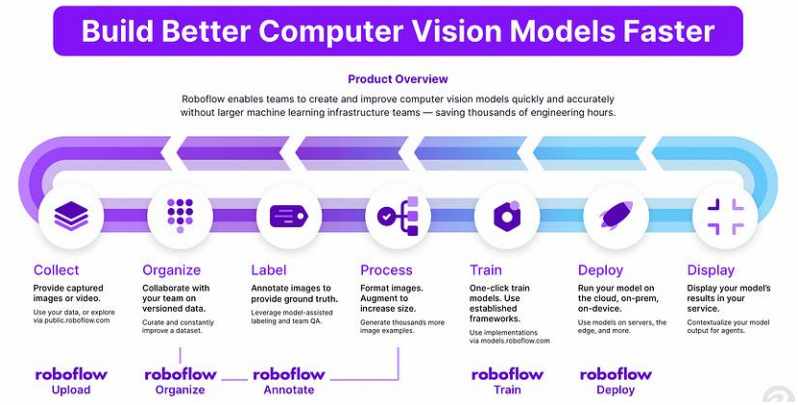


Figure: Roboflow Application

3. Roboflow-driven Preprocessing:
- Missing Data Handling: With Roboflow, identify any corrupted or unclear images. These may be discarded or flagged for manual review.
- Initial Feature Extraction: Retain key regions in images that contain the license plates, facilitating subsequent recognition.

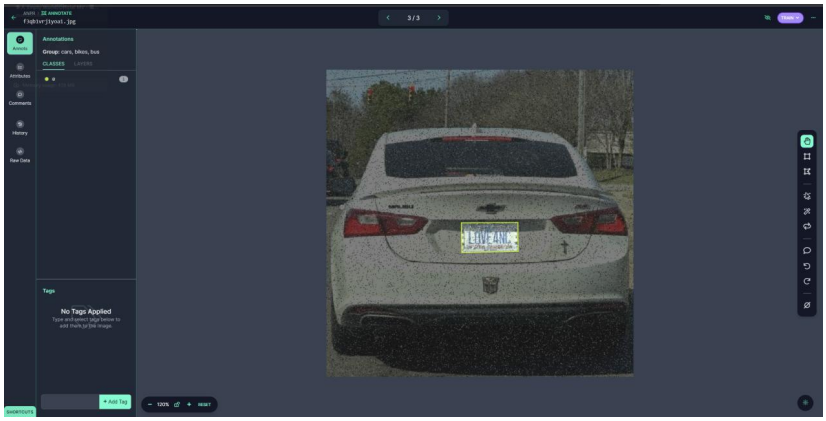


Figure: Creating the dataset from the Roboflow

4. Data Augmentation via Roboflow:
- Enhance the dataset's diversity with Roboflow's data augmentation capabilities, creating variations of existing images to simulate different conditions such as nighttime, rain, and different angles.

5. Dataset Split with Roboflow:
- Segregate the enriched dataset into training and testing subsets using Roboflow. The training subset will be used to train YOLOv8, and the testing subset to evaluate its accuracy.

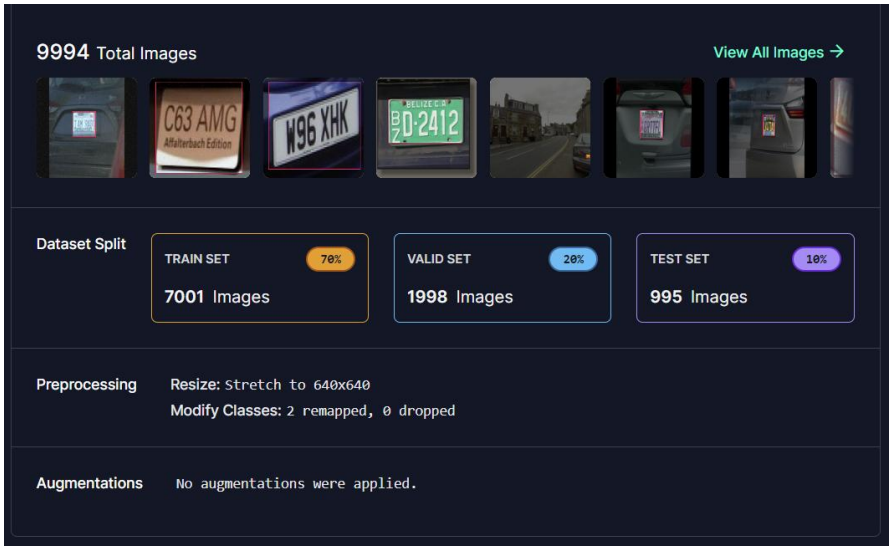


Figure: Spitting the dataset

6. Initialize YOLOv8 and Prep for Training:
- Opt for YOLOv8, renowned for its proficiency in real-time object detection. Its architecture, designed for pattern recognition in images, is particularly apt for identifying alphanumerics on license plates.

7. High-speed Model Training on Paperspace:
- To harness accelerated training speeds and efficient GPU utilization, deploy the dataset onto Paperspace. Training deep learning models on Paperspace's powerful GPU infrastructure ensures faster iterations, allowing for more extensive model tuning and optimization in a shorter time span.

8. Iterative Evaluation and Cross-validation:

- To foster model resilience and mitigate overfitting, conduct iterative training and validation of YOLOv8. This guarantees the model's capability to generalize to new, unseen data.

9. Predictive Model Candidate Generation:
- Post-training, the YOLOv8 model stands primed for license plate recognition.

10. Test Dataset Evaluation:
- Deploy the test dataset on the model. Consider metrics such as Intersection Over Union (IoU) to discern the model's accuracy in recognizing and isolating license plates in images. We can use Wandb for result visualization.

11. Finalize Predictive ANPR Model:
- Once satisfactory metrics are obtained, the YOLOv8 model is finalized for real-time deployment across Cambodia.

12. Real-time Application:
- The resultant model, fortified by Roboflow's preprocessing suite and trained on Paperspace's robust infrastructure, is now adept at recognizing license plates from fresh, real-time image feeds captured across the nation.

Architecture of the Artificial Intelligent model

1. System Architecture
- Hardware: We employ a 1080p Camera equipped with an AI model.
- Software: Our system runs on both Windows and Linux OS. It uses a Firebase database, algorithms driven by Python, and a website designed using Javascript.

2. Image Capture Device
For the demo, since real-time data detection isn't possible, we utilize a video captured in the United Kingdom at the specific location of “XX00 XXXX”.

3. Adjusting the Video
The video settings are adjusted based on vehicle speed or location to ensure optimal algorithm performance. For instance, faster vehicles require a lower frame rate to capture adequate data in a short time span. This also helps reduce computational load. For testing, we might employ a frame rate of, say, 40 frames per second.

```
while ret:
    frame_nmr += 1
    ret, frame = cap.read()
    if ret and frame_nmr % frame_skip == 0:
        print(f"Processing frame {frame_nmr}")  # Debug print
        results[frame_nmr] = {}
```

4. Vehicle Detection
We utilize cutting-edge technology, YOLOv8, to run our detection algorithms. YOLOv8 identifies vehicles in the present frame. Only vehicles with relevant class IDs are further

processed.

```python
x1, y1, x2, y2, score, class_id = license_plate
xcar1, ycar1, xcar2, ycar2, car_id = get_car(license_plate, track_ids)

if car_id != -1:
    # ...
    license_plate_crop = frame[int(y1):int(y2), int(x1): int(x2), :]
    license_plate_crop_gray = cv2.cvtColor(license_plate_crop, cv2.COLOR_BGR2GRAY)
    _, license_plate_crop_thresh = cv2.threshold(license_plate_crop_gray, 64, 255, cv2.THRESH_BINARY_INV)
    license_plate_text, license_plate_text_score = read_license_plate(license_plate_crop_thresh)

```

5. Vehicle Tracking

We employ the SORT algorithm (Simple Online and Real-time Tracking) to track vehicles across frames using the coordinates of bounding boxes provided by the YOLO model.

6. License Plate Association and Reading

Each identified license plate gets associated with a vehicle via the 'get_car' function (this function derives from vehicle tracking using the SORT algorithm). License plates are then cropped from the frame and processed to enhance readability. The 'read_license_plate' function recognizes the text on the license plate. This function uses the easyOCR library. However, for superior accuracy, we can design a specific OCR algorithm exclusive to Cambodian Number Plate Recognition. The font and location of area codes can be defined for detailed analysis.

.

```python
for license_plate in license_plates.boxes.data.tolist():
    x1, y1, x2, y2, score, class_id = license_plate
    xcar1, ycar1, xcar2, ycar2, car_id = get_car(license_plate, track_ids)

    if car_id != -1:
        # ...
        license_plate_crop = frame[int(y1):int(y2), int(x1): int(x2), :]
        license_plate_crop_gray = cv2.cvtColor(license_plate_crop, cv2.COLOR_BGR2GRAY)
        _, license_plate_crop_thresh = cv2.threshold(license_plate_crop_gray, 64, 255, cv2.THRESH_BINARY_INV)
        license_plate_text, license_plate_text_score = read_license_plate(license_plate_crop_thresh)

```

7. Data Saving and Uploading:

Detected car and license plate images are saved locally before being uploaded to Firebase. We chose Firebase for its user-friendly integration during the testing phase. However, for the actual project, considering the vast data queries, we'd recommend transitioning to platforms like Apache Spark. Subsequently, databases should migrate from Firebase to more robust platforms like Amazon S3 or Google Cloud Buckets. Finally, detected license plate texts and timestamps are stored in Firebase under 'users_detected'.

```python
# ...
car_image = frame[int(ycar1):int(ycar2), int(xcar1): int(xcar2), :]
car_image_filename = os.path.join(car_output_dir, f"car_{license_plate_text}.jpg")
cv2.imwrite(car_image_filename, car_image)
car_image_url = upload_to_firebase(car_image_filename, f"detected_cars/car_{license_plate_text}.jpg")
# ...
new_user = users_ref.push({
    'license_plate': license_plate_text,
    'timestamp': current_time
})
```

8. Flagged Data Identification:

Our script fetches a list of previously detected license plates from Firebase. It then compares this with the current database of license plates, flagging any matches.

```python
# Fetch data from Firebase
users_detected_data = users_ref.get()
users_database_ref = ref.child('users_database')
users_database_data = users_database_ref.get()

# Extract license plate values
detected_license_plates = {uid: details['license_plate'] for uid, details in users_detected_data.items()}
database_license_plates = {uid: entry['License_Plate'] for uid, entry in enumerate(users_database_data) if 'License_Plate' in entry}

# Find matching license plates and store the complete data
flagged_data = {}

for uid, license_plate in detected_license_plates.items():
    if license_plate in database_license_plates.values():
        flagged_data[uid] = users_detected_data[uid]
```

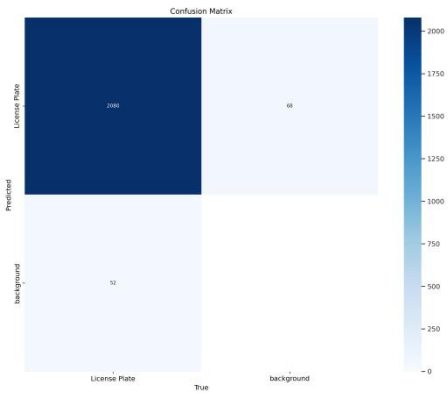
9. Flagged Data Uploading:

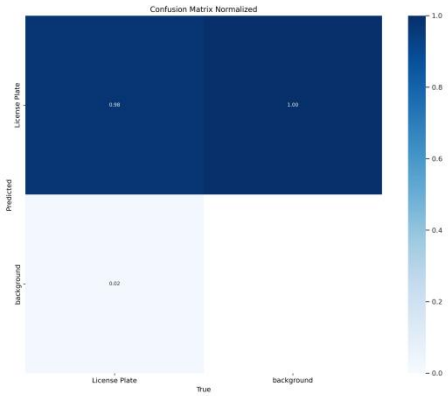
Details of flagged users are gathered, displayed on the console, and then uploaded to Firebase under 'flagged_details'.

```python
if flagged_data:
    flagged_ref = ref.child('flagged')
    flagged_ref.set(flagged_data)

# Collect details from users_database for flagged License_Plate and upload to Firebase
# ...
if flagged_details_data:
    flagged_details_ref = ref.child('flagged_details')
    flagged_details_ref.set(flagged_details_data)
```

Algorithm Performance Analysis:





Confusion Matrix:

- True Positives (TP): **2080**

This means 2080 times the model correctly predicted a license plate when there was indeed a license plate.

- True Negatives (TN): **68**

This means 68 times the model correctly predicted the background when there was actually a background.

- False Positives (FP): **52**

This means 52 times the model predicted a license plate when there was actually a background.

- False Negatives (FN): **68**

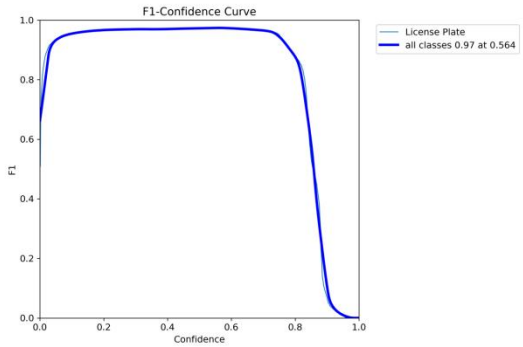
This means 68 times the model predicted the background when there was actually a license plate.

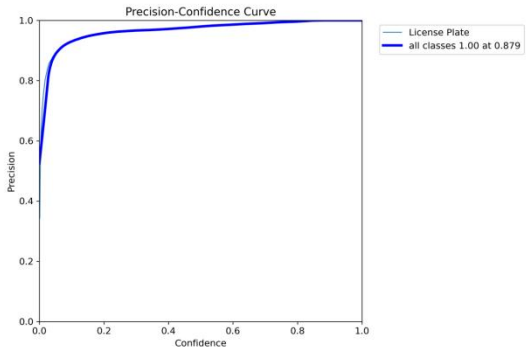
Confusion Matrix Normalized:

- TP: **0.98**
- TN: **1.00**
- FP: **0.02**
- FN: Not given directly, but it can be inferred as **0.02**

Analysis:

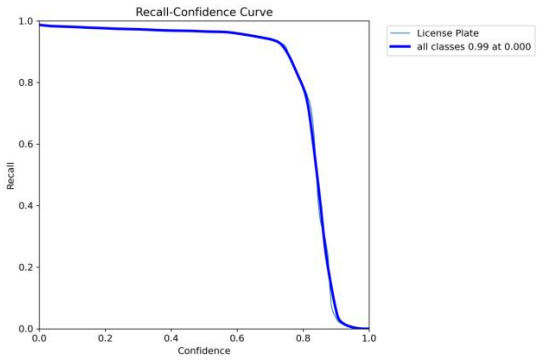
- Your model has a high True Positive rate of 0.98, meaning it's very effective at detecting license plates when they are present.
- The True Negative rate is 1.00, which indicates that when the model predicts "background," it is always correct.
- The False Positive rate is relatively low at 0.02, which means the model occasionally predicts license plates when they aren't there, but not often.
- The False Negative rate can be inferred to be 0.02, meaning the model sometimes misses license plates when they are present, but again, this happens rarely.

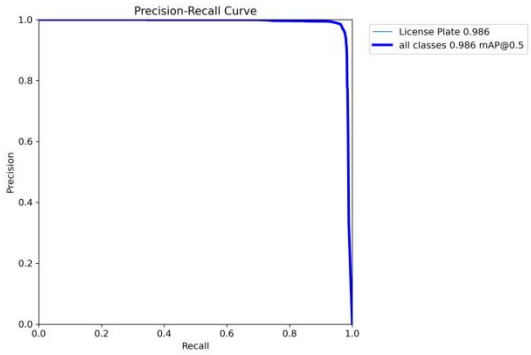




1. **F1-Confidence Curve**: This curve plots the F1 score against varying confidence thresholds. The F1 score is a measure that considers both precision and recall, offering a balance between the two. A score of 1 indicates perfect precision and recall, while a score of 0 indicates neither precision nor recall. From the curve, it seems that the model performs quite well for the "License Plate" class with the F1 score approaching 1 at a confidence level of around 0.564. This indicates that for detections with a confidence higher than 0.564, the model has an F1 score of 0.97 for detecting license plates.

2. **Precision-Confidence Curve**: This curve plots the precision against varying confidence thresholds. Precision is a measure of how many of the positive identifications were actually correct. In the provided curve, it can be observed that for the "License Plate" class, the model achieves a precision of 1 (perfect precision) at a confidence level of 0.879. This means that detections above this confidence level are all correct, at least according to the test data.





3. **Precision-Recall Curve**: This graph plots precision against recall for different threshold values. Precision measures the correctness of the positive predictions made by a model, while recall measures the proportion of actual positives that were correctly identified. The curve demonstrates how precision and recall trade off as the threshold changes. A model that performs perfectly would have a curve that goes straight up at recall=0 and then straight to the right at precision=1, forming a right angle. For the Precision-Recall Curve, the model has a mean Average Precision (mAP) of 0.986 at a threshold of 0.5 for license plate detection. This is a high score, indicating that the model performs well in both precision and recall for this class.

4. **Recall-Confidence Curve**: This graph plots recall against the confidence score, which is often the output of a classification model indicating the probability that a given instance belongs to the positive class. The curve shows how recall changes as the model's confidence threshold is varied. A model that is very confident in its predictions and also accurate would have a high recall at high confidence levels. For the Recall-Confidence

Curve, the model achieves a recall of 0.99 even at very low confidence levels (close to 0.000), suggesting that the model is very confident in its predictions and also achieves high recall.

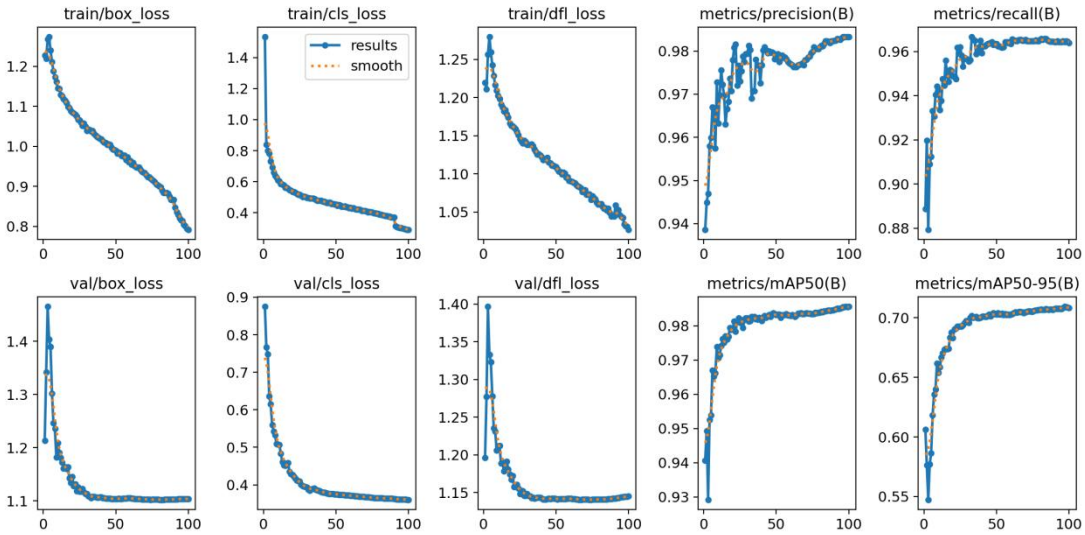


Figure: Training Result of the Model

Here's a quick analysis of each plot:

- **train/box_loss & val/box_loss:** Both training and validation box loss seem to decrease over time, indicating the model is learning. There's an initial spike in the validation loss, but it stabilizes afterward.

- **train/cls_loss & val/cls_loss:** The classification loss for both training and validation sets decrease as well, suggesting the model is classifying objects correctly as training progresses.

- **train/dfl_loss & val/dfl_loss:** This could represent another type of loss, perhaps related to the locations or dimensions of the detected boxes. It's also showing a decreasing trend, which is good.

- **metrics/precision(B) & metrics/recall(B):** Precision measures the accuracy of the positive predictions, while recall measures the fraction of positives that were correctly identified. Both metrics are trending upwards, which is promising. The fluctuations in precision might suggest some variance in the model's performance on different batches or epochs.

- **metrics/mAP50(B)**: mAP, or mean average precision, is a common metric in object detection tasks. It measures the average precision of the model at different recall levels. mAP at IoU (Intersection over Union) of 0.50 indicates the model's performance when a predicted box is considered correct if it has an IoU of at least 0.50 with a ground truth box. The trend is increasing, which suggests improving performance.

- **metrics/mAP50-95(B)**: This metric averages the mAP from IoU of 0.50 to 0.95 with a step of 0.05. It provides a more comprehensive view of the model's performance across different IoU thresholds. The trend is increasing, which is a positive sign.

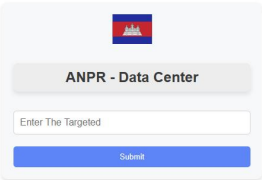
Main Application

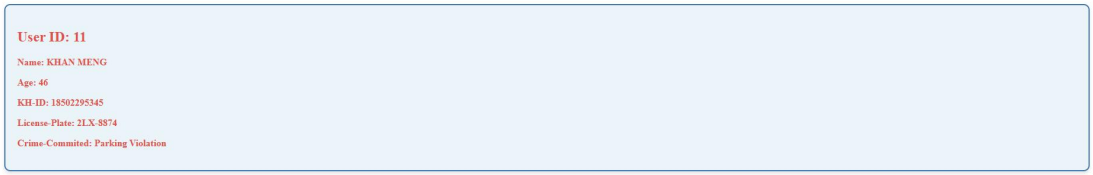
For the main application of the project, it included into three main routers, moreover Authentication also provided in this project:

1. /admin

Search and Query Functionality

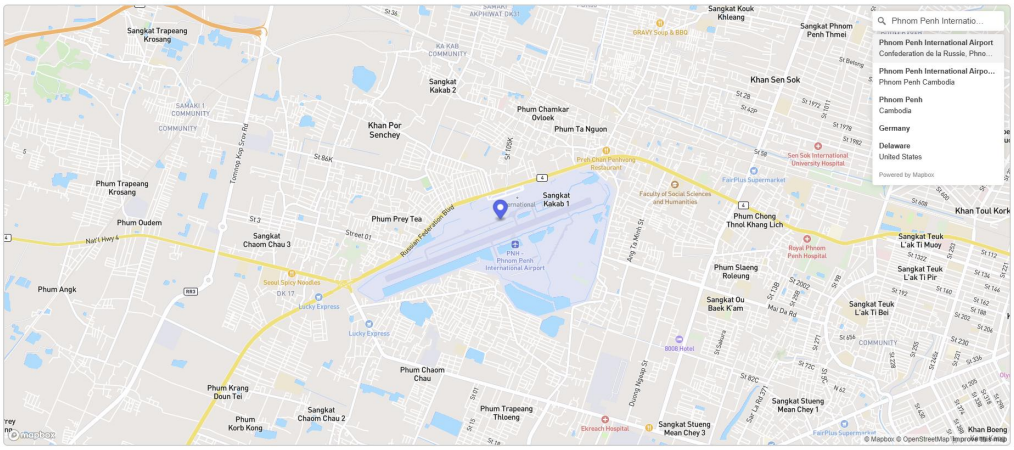
Purpose: This feature allows users to effortlessly search and extract data related to a particular vehicle or a set of vehicles based on specific criteria.
How to use: Simply enter your search criteria into the provided search bar. It could be based on vehicle registration, date of registration, vehicle type, or any other available parameter.
Results: Once you input your criteria and hit search, you'll be presented with a list of results matching your query. Each result will have detailed information regarding the vehicle.





Smart Camera Map View

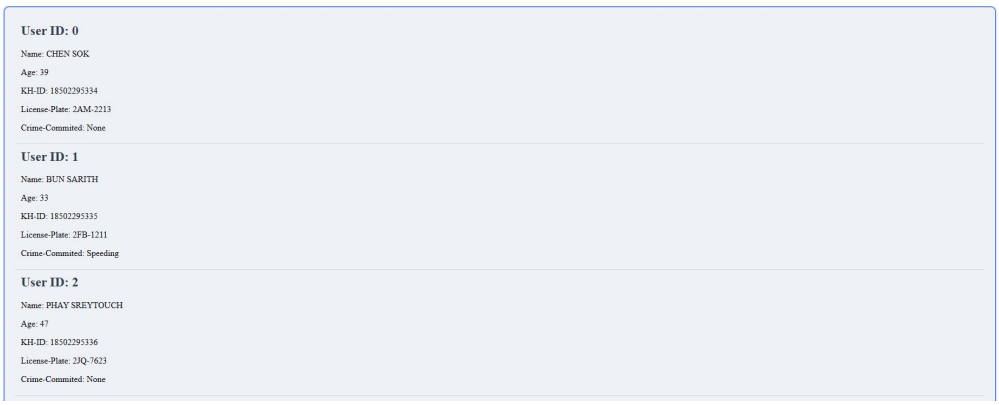
Purpose: To provide users with a geographical perspective, we've integrated a map that showcases the locations of all our smart cameras.
How to use: Below the search bar, you will find an interactive map. Zoom in or out, and click on camera icons to view more details about a specific location.
Benefit: This visual representation assists users in understanding the spread and coverage of our smart camera network. Whether you are trying to trace the route of a specific vehicle or understand where the data is coming from, this map is your key.



Vehicle Registration Data Display

Purpose: This section offers a comprehensive view of all vehicle registration data in our system.
Information Included: For each vehicle, details such as the vehicle's make and model, registration number, registration date, and any associated ANPR data will be available.

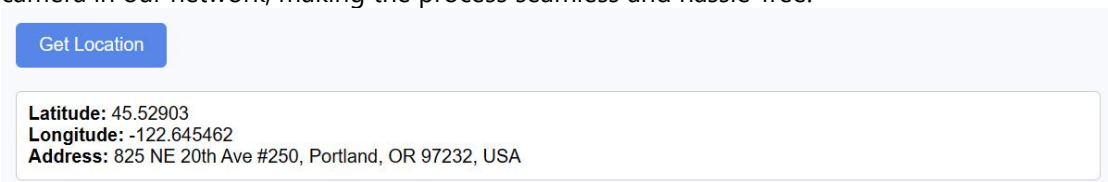
How to use: Browse through the list or use the search functionality to narrow down to specific vehicles. Click on any vehicle entry to view more in-depth details.



2. /detected

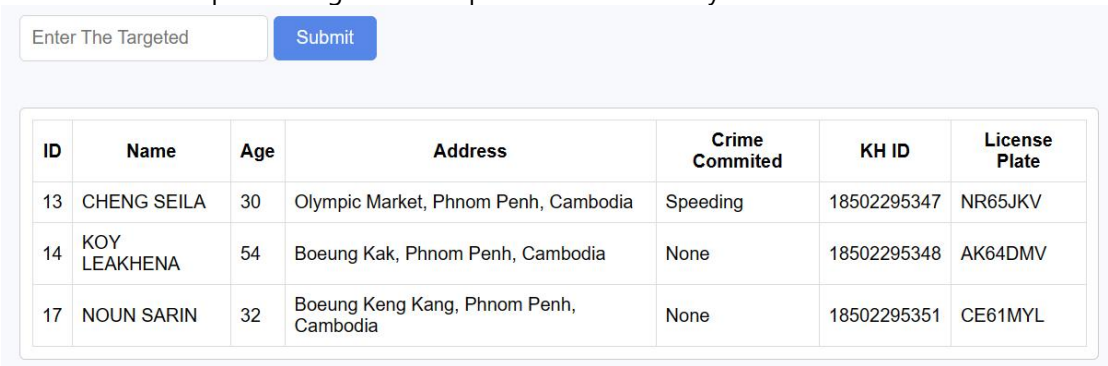
Quick Location Retrieval:

Purpose: Designed for speed and efficiency, this tool lets you swiftly pinpoint the exact location of any smart camera.
How to Use: While you can always retrieve locations using the camera's ID, with our integrated function, it's just a click away. You can immediately access any camera's location without the need for manual ID entry. This one-time built-in feature is applicable to every camera in our network, making the process seamless and hassle-free.



Data Submission & Real-time Alerts:

Purpose: A direct link to our Firebase Database ensures the smooth flow of crucial data. Whenever a target number plate is detected, you'll be the first to know!
How to Use: Use the provided form to submit details of your target number plates. Once entered, the system keeps a vigilant watch. When a match is detected, the application fetches the relevant data and showcases it in a structured table format. This table will display the detected number plate alongside other pertinent details for your review.



ID	Name	Age	Address	Crime Commited	KH ID	License Plate
13	CHENG SEILA	30	Olympic Market, Phnom Penh, Cambodia	Speeding	18502295347	NR65JKV
14	KOY LEAKHENA	54	Boeung Kak, Phnom Penh, Cambodia	None	18502295348	AK64DMV
17	NOUN SARIN	32	Boeung Keng Kang, Phnom Penh, Cambodia	None	18502295351	CE61MYL

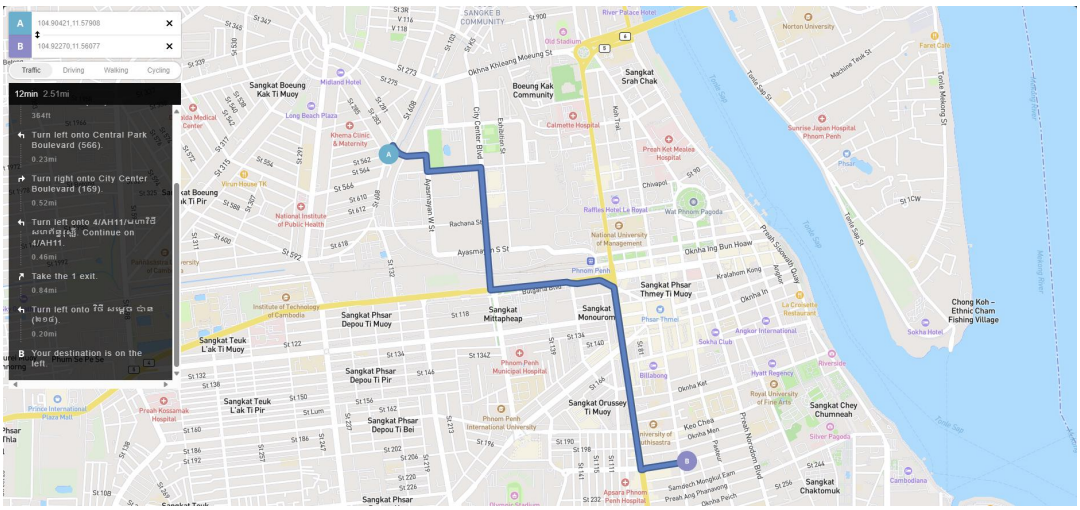
Photo Verification Feature:

Purpose: Accuracy matters! Our system provides real-time photos of vehicles to assist you in cross-verification, ensuring the utmost confidence in our detection process.
How to Use: Just below the data presentation table, you'll find photos of the detected vehicles along with their plate numbers. Especially valuable in rare instances where our detection might face challenges, this feature allows you to visually confirm the type, color, and model of any suspicious vehicles, enhancing the trustworthiness of our system.

Timestamp & Number Plate Updates:

Purpose: Keeping you updated is our priority. This feature ensures you have the latest data, right down to the second!
How to Use: As vehicles are detected, our system instantly updates the number plate details, paired with an accurate timestamp. This ensures you're always viewing the most recent and relevant vehicle data.

3. /maps



Immediate Response:
Detail: By determining the quickest route, police or security personnel can immediately pursue or reach a vehicle of interest. This is essential during time-sensitive situations, such as chasing a suspect or reaching a potential crime scene.

Benefit: Reduces the window of opportunity for a suspect to escape or commit further offenses.

Optimized Resource Allocation:
Detail: With a system that calculates the fastest distance, dispatchers can direct units based on their proximity and availability to an incident.
Benefit: This ensures that the closest available units respond, conserving resources and improving response times.

Improved Strategy and Planning:
Detail: A comprehensive map can show the locations of all ANPR cameras, helping the police and security personnel understand where a vehicle has been and predict where it might be headed.
Benefit: Allows for setting up roadblocks, strategic interception points, or rerouting patrols in anticipation.

Evidence and Accountability:
Detail: Accurate mapping and route calculation can provide detailed logs of police movements, response times, and actions taken.
Benefit: Such records are crucial for maintaining transparency, reviewing operations, and providing evidence if required in judicial proceedings.

Benefits of ANPR in Cambodia

1. Economic Impacts: One of the most tangible benefits of the ANPR system is its potential to significantly boost the nation's revenue. With accurate and timely identification of vehicles, authorities can ensure that no vehicle evades the requisite fines or vehicle taxes. Such strict compliance could lead to an estimated increase in revenue, possibly amounting to around $5 million annually.

2. Security Benefits: The ANPR system acts as a vigilant eye on the roads, offering quicker identification of suspicious or flagged vehicles. This immediate recognition aids law enforcement agencies in preemptive actions and crime prevention, ensuring safer streets and communities.

3. Traffic Management: The real-time data gathered by the ANPR system can be a game-changer for urban planning and traffic management. By understanding traffic patterns, peak times, and congestion zones, city planners can make informed decisions, leading to data-driven traffic planning, optimized traffic light timings, and overall reduced congestion.

4. Better Situation Awareness: ANPR provides authorities with an instant snapshot of the vehicular landscape. Whether it's detecting an unusual surge in traffic in a particular area or identifying areas with lower vehicle movement, the system offers a comprehensive view of the situation, allowing for proactive measures.

5. Faster and More Accurate Results: Manual methods of vehicle identification are not only time-consuming but also prone to errors. ANPR, on the other hand, guarantees rapid and precise recognition, ensuring that no vehicle goes undetected or misidentified.

6. Improved Investigations: For law enforcement agencies, ANPR can be an invaluable tool in investigations. Whether tracing a suspect's vehicle movement or identifying vehicles present at a crime scene, the system provides accurate data that can expedite case resolutions.

7. Increased Efficiency: Eliminating the need for manual checks and data entry, ANPR streamlines the entire process of vehicle identification and monitoring. This automation not only reduces the chances of human error but also ensures that resources are utilized more efficiently.
8. Enhanced Road Safety: With real-time alerts about vehicles that might be involved in reckless driving or have a history of traffic violations, traffic police can take immediate action, thereby promoting road safety.

10. **Support for Smart City Initiatives:** As Cambodia aims for urban development, ANPR can be an essential component of smart city solutions, integrating with other technologies to create a cohesive, technologically advanced urban environment.

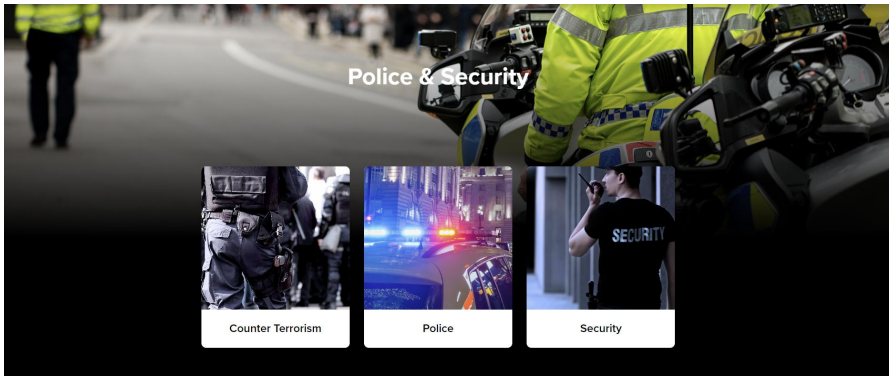


Figure: example of ANPR benefits

What we should Study more about Cambodia ANPR

Certainly! To effectively implement an ANPR system in Cambodia, especially considering its unique traffic and infrastructure challenges, a detailed study is crucial. Here's an expanded list of what needs to be studied:

1. Location Analysis
- Where to put the Camera: This involves conducting a traffic density analysis to identify high-traffic zones, intersections, entry and exit points of the city, and places with a high incidence of traffic violations.
- Strategic Points: Identify areas with higher security concerns, like government buildings, tourist attractions, and business districts.
- Residential vs. Commercial Areas: The flow of traffic varies, and understanding this can guide optimal camera placement.

2. Camera Positioning
- Camera Angle: The angle should be such that the camera can capture the license plates without any obstruction, especially during peak traffic hours.
- Height of the Camera: This is crucial to avoid obstructions like buses or trucks. It should also be out of easy reach to prevent tampering or vandalism.
- Orientation: Some roads may require cameras to be placed facing oncoming traffic, while others may benefit from side or overhead shots.

3. Camera Specifications:

- Type of the Camera: This involves deciding between options like infrared cameras for nighttime vision, high-definition cameras for clearer images, or pan-tilt-zoom cameras for wider coverage.
- Frame Rate: The camera should capture multiple frames per second to ensure no vehicle is missed during heavy traffic flow.
- Weatherproofing: Given the varied weather conditions in Cambodia, the cameras need to be resistant to heavy rains, dust, and heat.
- Vandal-proofing: Cameras should be sturdy enough to withstand tampering or deliberate damage attempts.

4. Data Management:
- Data to use: This entails understanding the kind of data the system will capture – apart from license plate numbers, this could include vehicle type, color, time-stamp, and more.
- Storage: Decide on local storage vs. cloud storage. Given the massive amount of data ANPR systems generate, robust and scalable storage solutions are essential.
- Data Retention: Establish policies on how long the data will be stored and when it will be purged.
- Privacy Measures: Ensure that data handling and storage are compliant with privacy laws and regulations, ensuring the public's trust.

5. Infrastructure and Integration:
- Power Backup: Continuous power supply is crucial. Study the power infrastructure and consider backup solutions like solar panels or generators.
- Integration with Other Systems: See how the ANPR can be integrated with existing traffic management systems, police databases, or other smart city solutions.

6. Public Awareness:
- Educate the Public: Before the rollout, awareness campaigns about ANPR's benefits and functionalities can help in gaining public trust and cooperation.

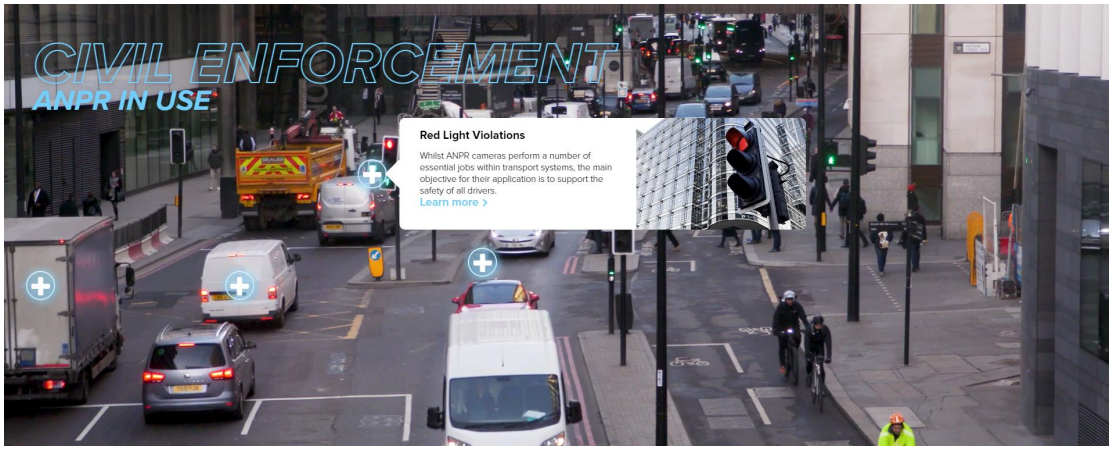
7. Maintenance:
- Regular checks and maintenance schedules to ensure cameras and other equipment are functioning optimally.

1. Data Collection:
- Good amount of data is require to train a good Artificial Intelligence model, at least 10,000 data is enough to feed into the training.

Conducting thorough research in these areas will help in the effective implementation and success of the ANPR system in Cambodia.

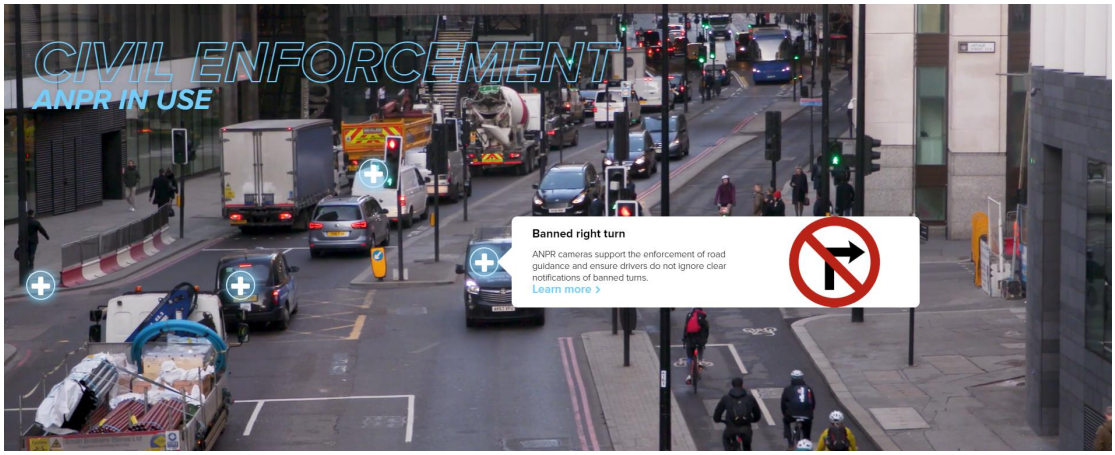
Enhanced Utility of ANPR in Cambodia for Improved Traffic Management and Safety:

1. Line at the Stop Violations:
- Detail: ANPR can monitor vehicles at intersections and detect those that cross designated stop lines. This is particularly important at traffic signal points or pedestrian crossings.
- Supporting Point: Such detection can deter drivers from making risky moves, reducing the chance of intersection-related accidents, especially those involving pedestrians.



2. Violations in Left or Right Turns:

- Detail: The system can be calibrated to understand the allowed turn directions at intersections and identify vehicles that violate these directives.
- Supporting Point: By actively monitoring and penalizing incorrect turns, traffic flow can be maintained, and the number of side-impact collisions can be reduced.



3. Driving Against Traffic:

- Detail: ANPR can spot vehicles moving in the wrong direction, especially on one-way streets or designated lanes.
- Supporting Point: This capability is crucial in preventing head-on collisions and maintaining the integrity of traffic flow, especially in busy city centers like Phnom Penh.

4. Toll Road Enforcement:

- Detail: ANPR can be used to automatically register vehicles passing through toll points, allowing for cashless and swift toll collection.
- Supporting Point: This reduces congestion at toll booths, provides a contactless payment method, and ensures all vehicles are charged appropriately.

5. Detecting Bad Driving Behaviors:

- Detail: Beyond license plate recognition, ANPR systems integrated with AI can detect erratic driving patterns, speeding, or other dangerous behaviors.
- Supporting Point: Such detections can be used to penalize habitual offenders and might act as a deterrent, promoting safer driving habits.

6. On-Screen Violation Alerts:

- Detail: Public display boards can be integrated with ANPR to show details of vehicles that violate traffic laws, serving as a real-time deterrent.
- Supporting Point: The public shaming might act as a strong deterrent, discouraging drivers from breaking the law, knowing they might be instantly displayed.

7. Immediate Reporting in Case of Robberies:
- Detail: If a crime like a street robbery occurs, victims or witnesses can quickly report the suspect's license plate. ANPR can then be used to track and locate the suspect's vehicle in real-time.
- Supporting Point: Such a system increases the efficiency of law enforcement response and may lead to quicker resolutions while acting as a deterrent for potential criminals aware of the surveillance.

Future Enhancement and Scalability of ANPR in Cambodia:

Cambodia, with its burgeoning urban centers and increasing vehicular population, stands at the threshold of a technological transformation. The Automatic Number Plate Recognition (ANPR) system, initially introduced as a mechanism to monitor and manage traffic, has the potential to evolve into a multipurpose tool with wider applications. Here's a look at the potential future enhancements and scalability of the ANPR system in Cambodia:

1. Advanced AI and Machine Learning Integration: As the AI domain continues to progress, ANPR systems can be enhanced to recognize not just license plates but also vehicle types, colors, and even detect anomalies such as damages. By integrating deep learning, the system can be trained to predict traffic bottlenecks, paving the way for preemptive traffic management.

2. Real-time Traffic Management: With ANPR data, traffic signals in cities like Phnom Penh could be adjusted in real-time based on traffic flow, reducing congestion and improving the overall driving experience.

3. Integration with Smart City Infrastructure: As Cambodia leans towards the development of smart cities, ANPR can integrate seamlessly with other urban technologies. This could include synchronization with smart traffic lights, smart parking solutions, and even public transportation systems for efficient route planning.

4. Extended Database Collaborations: The ANPR system can collaborate with other databases, such as insurance companies, to instantly verify the insurance status of a vehicle, or with service centers to check the maintenance history, ensuring vehicles on the road are safe and compliant.

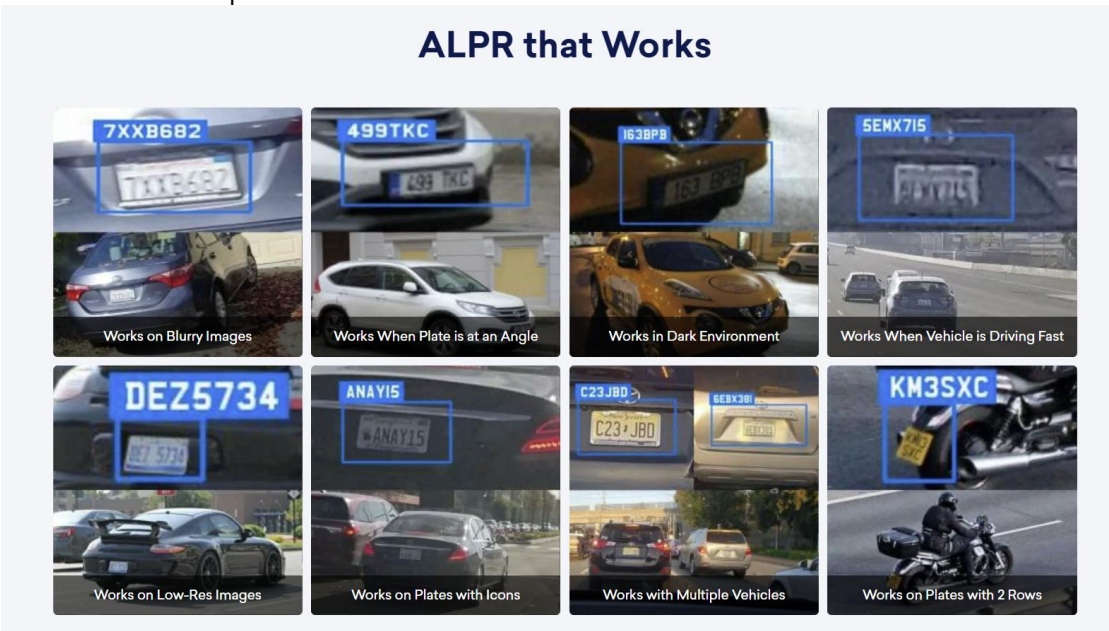
5. Enhanced Security Protocols: With advancements in cybersecurity, future iterations of ANPR can ensure encrypted data storage and transmission, assuring the public of their privacy and the safety of their data.

6. Environmentally Conscious Planning: Using the traffic flow data, city planners can identify areas with high vehicle emissions and implement eco-friendly initiatives, such as promoting electric vehicles or introducing more green spaces to counteract pollution.

7. Scalability Across Other Cities: While initial implementation might focus on major cities like Phnom Penh, the system can be scaled to other parts of Cambodia, ensuring nationwide traffic efficiency and security.

8. User-Friendly Public Interfaces: The future might see the development of mobile apps or platforms where citizens can access traffic data, pay fines, or even get updates about traffic conditions in real-time, promoting public participation and transparency.

9. Disaster Management Integration: In times of natural calamities or emergencies, the ANPR system can be employed to manage evacuations, identify stranded vehicles, or even to streamline rescue operations.



Conclusion

The transportation landscape of Cambodia stands at the cusp of a transformative phase. As urban challenges intensify, the implementation of Automated Number Plate Recognition (ANPR) offers a beacon of hope. Through features like photo verification, timestamps, and the meticulous mapping of vehicles, ANPR not only promises to enhance the management of vehicular movement but also to revolutionize the country's approach to transportation. Its potential economic, security, and traffic management benefits are vast and can lead to a safer, more efficient, and more sustainable urban environment. As we delve deeper into the realm of advanced AI and real-time monitoring, the scalability and adaptability of ANPR ensure that its relevance will only grow with time. The future of Cambodian transportation, with the integration of ANPR, looks promising, fostering a vision where roads are not just pathways but smart networks facilitating efficient, safe, and sustainable journeys for all.